



MAKE-UP EXAM

USN

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BMATS201

Second Semester B.E./B.Tech. Degree Examination, Nov./Dec. 2023 Mathematics – II for CSE Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. VTU Formula Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Evaluate $\int_{-c}^c \int_{-b}^b \int_{-a}^a (x^2 + y^2 + z^2) dz dy dx$	7	L2	CO1
	b.	Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ by changing into polar coordinates.	7	L2	CO1
	c.	Prove that $\beta(m, n) = \frac{\Gamma m - \Gamma n}{\Gamma m + n}$.	6	L2	CO1
OR					
Q.2	a.	Evaluate $\int_0^1 \int_x^{\sqrt{x}} xy dy dx$ by changing the order of integration.	7	L3	CO1
	b.	Prove that $\int_0^{\pi/2} \sqrt{\cot \theta} d\theta = \frac{\pi}{\sqrt{2}}$	7	L2	CO1
	c.	Using mathematical tools, write the code to find the area of an ellipse by double integration $A = 4 \int_0^a \int_0^{\frac{b}{a}\sqrt{a^2-x^2}} dy dx$	6	L3	CO5
Module – 2					
Q.3	a.	Find $\text{div } \vec{F}$ and $\text{curl } \vec{F}$, If $\vec{F} = \nabla(x^3 + y^3 + z^3 - 3xyz)$	7	L2	CO2
	b.	Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ Along the direction of the vector $(2i - j - 2K)$.	7	L2	CO2
	c.	Prove that the spherical coordinate system is orthogonal.	6	L3	CO2
OR					
Q.4	a.	Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$.	7	L3	CO2
	b.	Show that $\vec{F} = (y^2 - z^2 + 3yz - 2x)i + (3xz + 2xy)j + (3xy - 2xz + 2z)K$ is both solenoidal and irrotational	7	L2	CO2
	c.	Using the mathematical tools, write the codes to find the divergence of $\vec{F} = x^2yi + yz^2j + x^2zK$.	6	L3	CO5

Module – 3

Q.5	a.	Prove that the subset $W = \{(x, y, z) / x - 3y + 4z = 0\}$ of the vector space $\mathbb{R}^3$ is a subspace of $\mathbb{R}^3$.	7	L3	CO3
	b.	Determine whether the matrix $A = \begin{bmatrix} 3 & -1 \\ 1 & -2 \end{bmatrix}$ is a linear combination of $B = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$ and $D = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$ in the vector space M_{22} of 2×2 matrices.	7	L2	CO3
	c.	Find the linear transformation $T : V_2(\mathbb{R}) \rightarrow V_3(\mathbb{R})$ such that $T(1, 1) = (0, 1, 2)$, $T(-1, 1) = (2, 1, 0)$.	6	L2	CO3

OR

Q.6	a.	Determine whether the vectors $V_1 = (1, 2, 3)$, $V_2 = (3, 1, 7)$ and $V_3 = (2, 5, 8)$ are linearly dependent or linearly independent.	7	L2	CO3
	b.	Find the dimension and basis of the subspace spanned by the vectors $(2, 4, 2)$, $(1, -1, 0)$, $(1, 2, 1)$ and $(0, 2, 1)$ in $V_3(\mathbb{R})$	7	L2	CO3
	c.	Verify the rank-nullity theorem for the linear transformation $T : V_3(\mathbb{R}) \rightarrow V_3(\mathbb{R})$ defined by $T(x, y, z) = (x + 2y - z, y + z, x + y - 2z)$.	6	L2	CO3

Module – 4

Q.7	a.	Find the root of the equation $xe^x = 2$ that lies between 0 and 1. Using Regula- Falsi method. Carryout Four iterations. Correct to 3 – decimal places.	7	L2	CO4														
	b.	Use Newton's divided difference formula. Find $f(q)$, given the data : <table><tr><td>x</td><td>:</td><td>5</td><td>7</td><td>11</td><td>13</td><td>17</td></tr><tr><td>f(x)</td><td>:</td><td>150</td><td>392</td><td>1452</td><td>2366</td><td>5202</td></tr></table>	x	:	5	7	11	13	17	f(x)	:	150	392	1452	2366	5202	7	L3	CO4
x	:	5	7	11	13	17													
f(x)	:	150	392	1452	2366	5202													
	c.	Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by using Simpson's $\frac{1}{3}^{\text{rd}}$ rule taking 4 equal parts.	6	L3	CO4														

OR

Q.8	a.	Find the real root of the equation $3x - \cos x - 1 = 0$. Correct to 3-decimal places. Using Newton's Raphson method carryout 3 – iteration.	7	L2	CO4
	b.	Find $\tan(0.26)$ given that $\tan(0.10) = 0.1003$, $\tan(0.15) = 0.1511$, $\tan(0.20) = 0.2077$, $\tan(0.25) = 0.2553$, $\tan(0.30) = 0.3093$. Using Newton's Backward interpolation formula.	7	L2	CO4
	c.	Evaluate $\int_4^{5.2} \log x \, dx$ taking 6 equal parts. Using Simpson's $3/8^{\text{th}}$ rule.	6	L2	CO4

Module – 5

Q.9	a.	Employ Taylor's series method find y at $x = 0.1$ and 0.2 given that $\frac{dy}{dx} = 2y + 3e^x$; $y(0) = 0$. Up to fourth degree terms.	7	L2	CO4
	b.	Using Runge Kutta a method of forth order to find an approximate value of $y(0.2)$ given that $\frac{dy}{dx} = (x^2 + y)$ with $y(0) = 1$. Taking $h = 0.2$.	7	L2	CO4
	c.	Given $y' = (x - y^2)$ and the data $y(0) = 0$, $y(0.2) = 0.02$, $y(0.4) = 0.0795$, $y(0.6) = 0.1762$. Compute $y(0.8)$ by Milne's method.	6	L2	CO4
OR					
Q.10	a.	Using modified Euler method find $y(0.1)$ given that $\frac{dy}{dx} = (x + y)$, with $y(0) = 1$, Taking $h = 0.1$. Carryout 3-modification.	7	L2	CO4
	b.	Using Runge kutta method of fourth order, find the value of $y(0.2)$. Given that $\frac{dy}{dx} = \left(3x + \frac{y}{2}\right)$ with $y(0) = 1$. Taking $h = 0.2$.	7	L2	CO4
	c.	Using Mathematical tools, write the code solve the differential equation $\frac{dy}{dx} = 3e^x + 2y$ with $y(0) = 0$, using the Taylor's series method at $x = 0.1$ (0.1) 0.3.	6	L3	CO5

